

Finally, $f(E(\mathbf{U})) = f_N/f_D$, with

$$f_N = \frac{1}{G} \sum_{d=0}^1 (-1)^d \sum_{t \in \mathcal{T}_d} \sum_g N_{g,t} \left[E(\mathbb{1}\{D_{g,t} = 1-d, D_{g,t-1} = d\}(Y_{g,t} - Y_{g,t-1})) \right. \\ \left. - \frac{E(P_{1-d,d,t})}{E(P_{d,d,t})} E(\mathbb{1}\{D_{g,t} = D_{g,t-1} = d\}(Y_{g,t} - Y_{g,t-1})) \right] \\ f_D = \sum_{t \geq 2} E[P_{1,0,t} + P_{0,1,t}].$$

Reasoning as in the proof of Theorem 3 (see in particular Equations (20)-(21)) and noting that if $t \notin \mathcal{T}_d$, then, by Assumption S10 $\Pr(D_{g,t} = 1-d, D_{g,t-1} = d) = 0$, we get $f_N = E(T^S)$. Moreover, $f_D = E(P^S)$. Hence,

$$\sqrt{G} \left(\text{DID}_M - \frac{E(T^S)}{E(P^S)} \right) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \quad \text{with } \sigma^2 = J_f(\mathbf{U}^\infty) \Sigma J_f(\mathbf{U}^\infty)'. \quad (38)$$

Convergence to 0 of $\sqrt{G} (E(T^S)/E(P^S) - \delta^S)$

Let us define $\mathbf{D}_G = (D_{g,t})_{(g,t): g \leq G, t=1 \dots T}$, $\tilde{T} = E[T^S | \mathbf{D}_G]$ and $I = \mathbb{1}\{|P^S - E(P^S)| < \varepsilon_G\}$, where $\varepsilon_G > 0$ will be specified below. By Assumption S10, $\lim_{G \rightarrow \infty} E(P^S) > 0$. Thus, it suffices to prove that $\sqrt{G} (E(P^S)\delta^S - E(T^S)) \rightarrow 0$. Because $\delta^S = E[T^S/P^S]$, we have

$$E(P^S)\delta^S - E(T^S) = E \left[\tilde{T}(E(P^S) - P^S)/P^S \right] \\ = E \left[\tilde{T}I(E(P^S) - P^S)/P^S \right] + E \left[\tilde{T}(1-I)(E(P^S) - P^S)/P^S \right]. \quad (39)$$

First, consider the second term on the right-hand side. By applying twice the Cauchy-Schwarz inequality, we get

$$\sqrt{G} |E[\tilde{T}(1-I)(E(P^S) - P^S)/P^S]| \leq \sqrt{G} V(P^S)^{1/2} \left[E[(\tilde{T}/P^S)^4] \Pr(I=0) \right]^{1/4}$$

By Assumption S9, $\sqrt{G} V(P^S)^{1/2}$ converges towards a finite limit. Thus, it suffices to show that the term into brackets tends to 0. To this end, note first that

$$E[T^S | \mathbf{D}_G] = \frac{1}{G} \sum_{(g,t): t \geq 2} N_{g,t} \mathbb{1}\{D_{g,t} \neq D_{g,t-1}\} E[Y_{g,t}(1) - Y_{g,t}(0) | \mathbf{D}_G].$$

Now, let $A_{g,t} = N_{g,t} \mathbb{1}\{D_{g,t} \neq D_{g,t-1}\}$ and $B_{g,t} = E[Y_{g,t}(1) - Y_{g,t}(0) | \mathbf{D}_G]$. By Assumption S9 and Jensen's inequality, $\sup_{(g,t): g \leq G} E[|B_{g,t}|^4] < +\infty$. Then

$$\left(\frac{\tilde{T}}{P^S} \right)^4 \leq \left(\frac{\sum_{g,t} A_{g,t} |B_{g,t}|}{\sum_{g,t} A_{g,t}} \right)^4 \\ \leq \left(\max_{g,t} |B_{g,t}|^4 \right) \\ \leq \sum_{g,t} |B_{g,t}|^4.$$

Hence, $E[(\tilde{T}/P^S)^4] \Pr(I = 0) \leq K_1 G \Pr(I = 0)$ for some constant $K_1 > 0$. Moreover, by Hoeffding's inequality,

$$\Pr(I = 0) \leq 2 \exp \left(\frac{-2G^2 \varepsilon_G^2}{\sum_g N_{g,\cdot}^2} \right).$$

By Assumption S9, there exists $c > 0$ such that for all G , $1/G \sum_{g=1}^G N_{g,\cdot}^2 < 2c$. Then,

$$\frac{-2G^2 \varepsilon_G^2}{\sum_g N_{g,\cdot}^2} \leq -\frac{G \varepsilon_G^2}{c}.$$

Let $\varepsilon_G = (C \ln(G)/G)^{1/2}$, for some $C > c$. Then, by what precedes, $G \Pr(I = 0) \rightarrow 0$ and the second term of the right-hand side of (39) tends to zero.

Now, let us move to the first term of the right-hand side of (39). We have

$$\begin{aligned} & |E[\tilde{T}I(E(P^S) - P^S)/P^S]| \\ &= \left| E[I(\tilde{T} - E(\tilde{T}))(E(P^S) - P^S)/P^S] + E(\tilde{T}) (E(P^S)E(I/P^S) - E(I)) \right| \\ &\leq E(P^S) V(\tilde{T})^{1/2} \left(E[I(1/P^S - 1/E(P^S))^2] \right)^{1/2} \\ &\quad + |E(\tilde{T})| E(P^S) |E(I/P^S) - E(I)/E(P^S)|. \end{aligned} \tag{40}$$

We now prove that both terms on the right-hand side tend to zero. First, by Taylor expansions of $x \mapsto 1/x$ around $E(P^S)$, there exist (P_1^S, P_2^S) in the interval between P^S and $E(P^S)$ such that

$$\frac{1}{P^S} = \frac{1}{E(P^S)} - \frac{P^S - E(P^S)}{P_1^{S2}}, \tag{41}$$

$$\frac{1}{P^S} = \frac{1}{E(P^S)} - \frac{P^S - E(P^S)}{E(P^S)^2} + \frac{(P^S - E(P^S))^2}{P_2^{S3}}. \tag{42}$$

When $I = 1$, $|P_1^S - E(P^S)| < \varepsilon_G$ and $|P_2^S - E(P^S)| < \varepsilon_G$. Recall also that $\lim_G E(P^S) > 0$ and $\varepsilon_G \rightarrow 0$. Then, in view of (41) and by Assumption S9,

$$E[I(1/P^S - 1/E(P^S))^2] \leq \frac{V(P^S)}{(E(P^S) - \varepsilon_G)^4} \rightarrow 0.$$

Moreover, by definition of $\tilde{T}$, $GV(\tilde{T}) \leq GV(T^S)$, and the latter is bounded by Assumption S9. Therefore, the first term of the right-hand side of (40) tends to 0.

Now, multiplying (42) by I and taking the expectation on both sides, we obtain:

$$\begin{aligned} \sqrt{G} |E(I/P^S) - E(I)/E(P^S)| &\leq \sqrt{G} \left| \frac{E[I(P^S - E(P^S))]}{E(P^S)^2} \right| + \frac{\sqrt{G} V(P^S)}{(E(P^S) - \varepsilon_G)^3} \\ &\leq \left| \frac{[G \Pr(I = 0) V(P^S)]^{1/2}}{E(P^S)^2} \right| + \frac{\sqrt{G} V(P^S)}{(E(P^S) - \varepsilon_G)^3} \\ &\rightarrow 0. \end{aligned}$$

2. Validity of the confidence intervals

There exists $\bar{\sigma}_G^2$ such that $\sum_g \hat{\psi}_g^2/G - \bar{\sigma}_G^2 \xrightarrow{\mathbb{P}} 0$, with $\liminf_G \bar{\sigma}_G^2 \geq \sigma^2$.

Let $Q_{d,d',t}^\infty = \lim_{G \rightarrow \infty} E[Q_{d,d',t}]$ and $\boldsymbol{\lambda}_g$ be the column vector such that $\mathbf{U} = \sum_{g=1}^G \boldsymbol{\lambda}_g/G$. Some tedious algebra show that $J_f(\mathbf{U}^\infty) \times \boldsymbol{\lambda}_g = \psi_g$, with

$$\psi_g = \frac{1}{P_S^\infty} \sum_{t>1} N_{g,t} [\mathbf{1}\{D_{g,t} \neq D_{g,t-1}\}(Y_{g,t} - Y_{g,t-1} - f(\mathbf{U}^\infty)) - \psi_{g,t}^B]$$

and

$$\begin{aligned} \psi_{g,t}^B = & \frac{1}{P_{0,0,t}^\infty} \left[\mathbf{1}\{D_{g,t} > D_{g,t-1}\} Q_{0,0,t}^\infty + P_{1,0,t}^\infty \mathbf{1}\{D_{g,t} = D_{g,t-1} = 0\} \left(Y_{g,t} - Y_{g,t-1} - \frac{Q_{0,0,t}^\infty}{P_{0,0,t}^\infty} \right) \right] \\ & + \frac{1}{P_{1,1,t}^\infty} \left[\mathbf{1}\{D_{g,t} < D_{g,t-1}\} Q_{1,1,t}^\infty + P_{0,1,t}^\infty \mathbf{1}\{D_{g,t} = D_{g,t-1} = 1\} \left(Y_{g,t} - Y_{g,t-1} - \frac{Q_{1,1,t}^\infty}{P_{1,1,t}^\infty} \right) \right]. \end{aligned}$$

Hence, in view of (37)-(38), $\sigma^2 = \lim_G \sum_g V(\psi_g)/G$. Next, we show that $\sum_g \hat{\psi}_g^2/G$ is asymptotically larger than σ^2 . For that purpose, let $\bar{\sigma}_G^2 = \sum_g E(\psi_g^2)/G$ and remark that $\hat{\psi}_g = J_f(\mathbf{U}) \times \boldsymbol{\lambda}_g$. Then

$$\frac{1}{G} \sum_g \hat{\psi}_g^2 - \bar{\sigma}_G^2 = \frac{1}{G} \sum_g [\hat{\psi}_g^2 - \psi_g^2] + \left[\frac{1}{G} \sum_g \psi_g^2 - \bar{\sigma}_G^2 \right]. \quad (43)$$

Let $\lambda_{k,g}$ denote the k th coordinate of $\boldsymbol{\lambda}_g$. Assumption S9 ensures that

$$\sup_{k,g} E[\lambda_{k,g}^4] < +\infty. \quad (44)$$

Thus, we also have $\sup_g E(\psi_g^2) < +\infty$. Therefore, by the weak law of large numbers, the second term on the right-hand side of (43) converges to 0. Next,

$$\frac{1}{G} \sum_g [\hat{\psi}_g^2 - \psi_g^2] = (J_f(\mathbf{U}) - J_f(\mathbf{U}^\infty)) \left[\frac{1}{G} \sum_g \boldsymbol{\lambda}_g \boldsymbol{\lambda}_g' \right] (J_f(\mathbf{U}) + J_f(\mathbf{U}^\infty))'. \quad (45)$$

By (44) again and the weak law of large numbers, $\mathbf{U} \xrightarrow{\mathbb{P}} \mathbf{U}^\infty$. Moreover, f is continuously differentiable in a vicinity of $\mathbf{U}^\infty$. Thus, by the continuous mapping theorem, $J_f(\mathbf{U}) - J_f(\mathbf{U}^\infty) \xrightarrow{\mathbb{P}} 0$. By (44) once again and the weak law of large numbers,

$$\frac{1}{G} \sum_g (\boldsymbol{\lambda}_g \boldsymbol{\lambda}_g' - E[\boldsymbol{\lambda}_g \boldsymbol{\lambda}_g']) \xrightarrow{\mathbb{P}} 0.$$

Moreover, by the Cauchy-Schwarz inequality and (44), we have, for all k, ℓ ,

$$\left| \frac{1}{G} \sum_g E[\lambda_{k,g} \lambda_{\ell,g}] \right| \leq \sup_{k,\ell,g} E[|\lambda_{k,g} \lambda_{\ell,g}|] \leq \sup_{k,\ell,g} E[|\lambda_{k,g}|^2] < +\infty.$$

As a result, $\sum_g \boldsymbol{\lambda}_g \boldsymbol{\lambda}_g'/G = O_P(1)$. Finally, because $J_f(\mathbf{U})$ converges in probability, $J_f(\mathbf{U}) + J_f(\mathbf{U}^\infty) = O_P(1)$. Thus, in view of (45), the first term on the right-hand side of (43) converges in probability to 0. Hence, we have proven that $\sum_g \hat{\psi}_g^2/G - \bar{\sigma}_G^2 \xrightarrow{\mathbb{P}} 0$. Finally, $E(\psi_g^2) \geq V(\psi_g)$ and thus $\bar{\sigma}_G^2 - \sum_g V(\psi_g)/G \geq 0$. Therefore, $\liminf_G \bar{\sigma}_G^2 \geq \sigma^2$.

$\text{CI}_{1-\alpha}(\delta^S)$ is asymptotically conservative.

By (38), the convergence to 0 of $\sqrt{G} (E(T^S)/E(P^S) - \delta^S)$, (37), and since $\psi_g = J_f(\mathbf{U}^\infty) \times \boldsymbol{\lambda}_g$,

$$\sqrt{G} \frac{\text{DID}_M - \delta^S}{\hat{\sigma}} = \frac{\sigma}{\bar{\sigma}_G} \left[\frac{\bar{\sigma}_G}{\hat{\sigma}} \frac{\frac{1}{\sqrt{G}} \sum_g \psi_g}{\sigma} + o_P(1) \right].$$

Let Z_G denote the term into brackets. By the first step above and Slutsky's lemma $Z_G \xrightarrow{d} \mathcal{N}(0, 1)$. Fix $\eta > 0$. Because $\liminf_G \bar{\sigma}_G^2 \geq \sigma^2$, there exists G_0 such that for every $G \geq G_0$,

$$\sqrt{G} \left| \frac{\text{DID}_M - \delta^S}{\hat{\sigma}} \right| \leq (1 + \eta) |Z_G|.$$

Then, letting Φ denote the cdf of the standard normal distribution, we get

$$\limsup_G \Pr(\delta^S \in \text{CI}_{1-\alpha}) \geq \Phi\left(\frac{z_{1-\alpha/2}}{1+\eta}\right) - \Phi\left(\frac{z_{\alpha/2}}{1+\eta}\right).$$

The result follows by letting η tend to zero.